
Predicting Gene Expression with Methylation-GEX Relationships Across Promoter Regions with an HMM

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Abstract

DNA methylation generally has an inverse relationship with gene expression levels in the promoter region of a gene. To quantify this relationship, existing approaches have used LASSO, which treats each CpG site as an independent feature of a gene, and thus independent predictor of gene expression, but this linear regression method does not perform very well. However, a Hidden Markov Model can model the spatial relationships seen between CpG sites instead of assuming independence. The Hidden Markov Model outperformed a baseline LASSO model in predicting gene expression, demonstrating that the Hidden Markov Model's ability to assign states to CpG sites based on methylation levels and methylation-gene expression regressions allowed it to more accurately predict gene expression levels.

Keywords methylation, gene expression, Hidden Markov Model, LASSO

1. Introduction

DNA methylation is an epigenetic modification that downregulates or silences gene expression (Wiltshire, 2024). Cytosines are the most commonly methylated base, in regions called CpG sites. These sites are often found in CpG islands, or GC-rich regions of the genome that contain many such sites. CpG islands are commonly found in the promoter regions of genes, contributing to the direct negative relationship between methylation and gene expression (Ioshikhes & Zhang, 2000).

There already exist some tools to model methylation states. For example, DMCHMM is a Hidden Markov Model (HMM) that can identify differentially methylated CpG sites across samples (Shokoohi et al., 2019). LASSO, a linear regression method, has also been used to predict gene expression levels based on methylation, where each CpG site contributes independently to the model (Zhong et al., 2019).

DMCHMM demonstrates that HMMs are good at modeling the spatial structure of methylation across genes. Therefore, a natural extension of this framework is to utilize it to identify the link between methylation levels and gene expression, taking advantage of the HMM's ability to identify spatial structure across a sequence of CpG sites. This project aims to implement this architecture and compare its performance to a baseline LASSO model to predict gene expression.

2. Methods

Analyses were conducted using R version 4.5.2 (R Core Team, 2025).

2.1. Initial Analysis

Initially, WGBS and RNA-seq data were downloaded from ENCODE, filtering for fibroblasts to use data from a consistent

cell type. This step is crucial, since different cell types will have different methylation patterns and different gene expression profiles, so ensuring that there is no variation here means the results are comparable. However, this data did not produce good results when performing baseline analyses, suggesting that the dataset was not well-suited for this task.

Performing LASSO over the whole set of genes, as was done in Zhong et al., produced no signal, as seen in Figure 1.

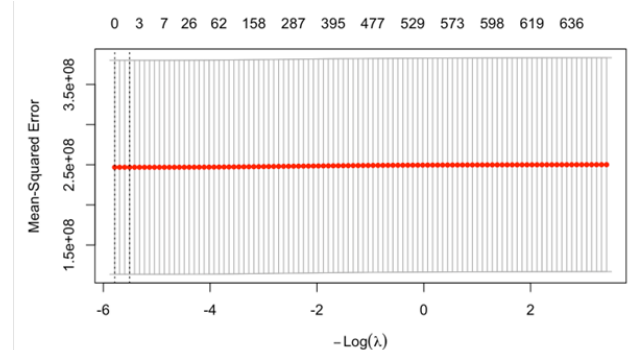


Figure 1 LASSO Results from Initial Dataset ($R^2 = 0.0005441$)

A simple correlation between methylation percentage and gene expression produced a Pearson correlation value very close to zero ($r = -0.008973$ for all genes, $r = -0.01176$ for expressed genes only), suggesting that although the direction of the correlation matched biological intuition – as methylation level increases, gene expression decreases – the strength of the correlation is practically nonexistent.

This mismatch with the LASSO results from Zhong et al., combined with the fact that each CpG-gene pair only had one sample, meaning that there was no variance between samples,

suggested that running an HMM on this data would not give good results. Therefore, a more appropriate data set was chosen.

2.2. Data Acquisition and Preprocessing

Data was downloaded from the colon cancer repository from The Cancer Genome Atlas (TCGA-COAD), consisting of methylation beta values, RNA-seq gene expression, and phenotype data. Data was filtered to include primary tumor samples only to reduce noise and keep a consistent cell type profile, as this excludes metastasized cells. Then, both methylation and RNA-seq data were filtered to include samples with genomic locations only in promoter regions using the `GenomicRanges` package, and the methylation and RNA-seq data were merged (Lawrence et al., 2013).

The final data set consisted of 137847 CpG sites across 20886 genes, consisting of 159947 CpG-gene pairs total. Before performing any other analysis, a simple correlation between methylation and gene expression was also visualized.

2.3. LASSO Analysis

LASSO was performed for each of the 1000 genes with the most CpG sites using the `glmnet` package (Friedman et al., 2010; Tay et al., 2023). The data was first transformed so it had one row per sample and one column per CpG site, then split into a training set (80%) and a test set (20%), where the input features were all the methylation columns and the output was the gene expression values. If any gene had less than 100 samples, it was dropped from the analysis. For each of the remaining genes, a model was fit with `cv.glmnet` using 10-fold cross-validation and performing feature selection, and this model was then stored. Then, the model for each gene was used on the test set, where the best value of λ was extracted and gene expression was predicted, then the R^2 value was calculated per gene. The overall results and the results for the best and worst genes were then analyzed.

2.4. HMM Analysis

For the HMM, the same 1000 target genes that LASSO was performed across were used to make a reliable comparison, and the train/test split was reused with the same samples in each split as in LASSO. The `depmixS4` package was used to create the HMM (Visser & Speekenbrink, 2010). Only the genes with at least ten CpG sites, with at least 100 samples, and for whom the HMM successfully converged were retained when fitting the HMM to ensure that there was enough data for a meaningful traversal. The HMM was structured to have three hidden states with parameters for a methylation distribution (μ , σ^2) and regression coefficients (β_0 , β_1) for predicting gene expression, all of which were learned by the Baum-Welch algorithm. Each of the three states had two emissions: a methylation value, which was logit-transformed to convert the beta values to values drawn from a Gaussian distribution; and a gene expression value, which was log-transformed. The HMM is visualized in Figure 2.

Then, for the training set, the following regression model was fitted for each state:

$$\log(GEX) = \beta_0 + \beta_1 \times \text{logit(methylation)}$$

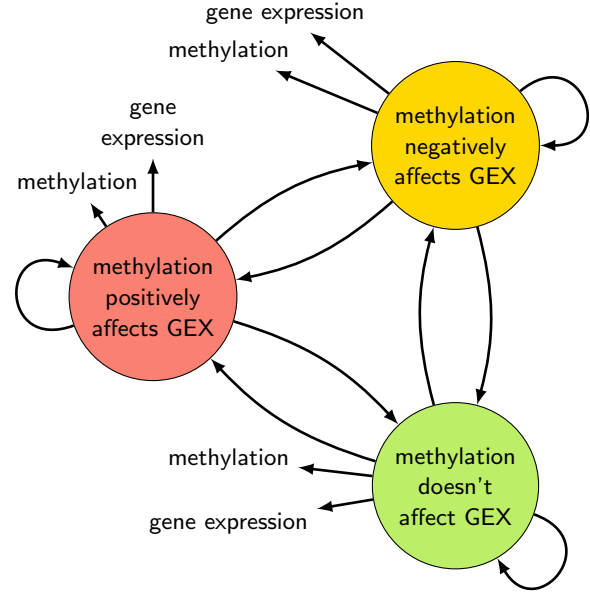


Figure 2 State diagram for HMM

Any genes that produced models that failed to converge were dropped.

For the test set, again, the same 20% of samples were used as in the LASSO analysis. The Viterbi algorithm was used so the HMM can assign a state to each CpG site based on its methylation value and its methylation-expression relationship. Then, the proportion of CpG sites in each state was calculated, and the predicted gene expression was calculated from a linear model of those proportions. Lastly, the R^2 was calculated per gene and stored.

3. Results

The simple correlation between methylation and gene expression produced a moderate negative correlation, which is an encouraging result since it's biologically accurate to the promoter region of a gene: when methylation is high, gene expression is low and vice versa. This data also shows a much stronger relationship than the original data set, indicating that the new data is more appropriate for this problem.

3.1. LASSO Results

The percentage of genes with R^2 greater than certain thresholds were calculated: 9.95% of genes had $R^2 > 0.3$, 2.81% of genes had $R^2 > 0.5$, and no genes had $R^2 > 0.8$. Since less than ten percent of genes had at least 30% of the variation in gene expression explained by variation in methylation, it is clear that LASSO does not capture the link between methylation and gene expression very well.

The results for the best-performing gene from LASSO are shown in Figure 4.

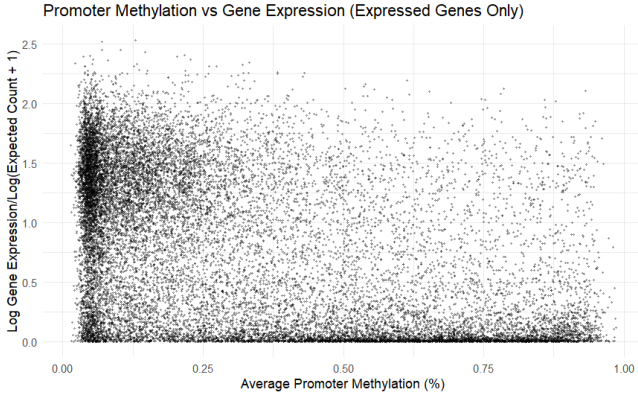


Figure 3 Correlation between methylation and gene expression ($r = -0.4328$ for all genes, $r = -0.431$ for expressed genes only)

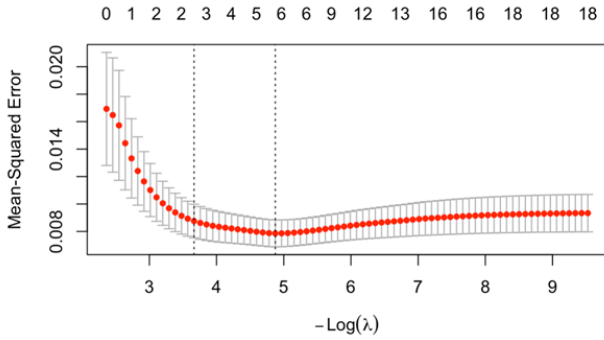


Figure 4 LASSO signal for the best gene (highest R^2)

3.2. HMM Results

The HMM was able to successfully fit models to 399 genes, after filtering out genes that did not meet the criteria listed or that had models that did not converge (i.e. the HMM was not able to identify a relationship between methylation and gene expression). When looking at the percentage of genes with R^2 greater than certain thresholds, similar to LASSO, the HMM produced the following results: 67.34% of genes had $R^2 > 0.3$, 56.53% of genes had $R^2 > 0.5$, and 8.29% of genes had $R^2 > 0.8$. As all of these percentages are higher than those produced by LASSO, the HMM clearly performs better than LASSO in predicting gene expression from methylation.

The emission parameters learned by the HMM for the best gene, or the gene for which the HMM produced the highest R^2 value, are summarized in Table 1. While HMM outperformed LASSO in predicting gene expression, it performed poorly in recognizing a distinct relationship between methylation and gene expression, with expression slopes of 0.000571 for state 1, -0.050789 for state 2, and -0.014226 for state 3, all very weak magnitudes. Meanwhile, there is slightly more variability in methylation beta mean values across the three states: 0.417 for state 1, 0.103 for state 2, and 0.349 for state 3. It is evident that the HMM assignments of states to CpG sites were less reliant on methylation-gene expression relationship and more on methylation value alone.

Table 1. Summary of parameters learned by HMM

State	Methylation beta mean (μ)	Expression intercept (β_0)	Expression slope (β_1)
1	0.417	0.0294	0.000571
2	0.103	1.8850	-0.050789
3	0.349	0.5876	-0.014226

From these parameters, the state descriptions were rewritten to better match the magnitude and direction of the learned expression slopes that were observed. States 1, 2, and 3 were redefined to be 'methylation doesn't affect GEX,' 'methylation negatively affects GEX,' and 'methylation slightly negatively affects GEX,' respectively. These changes are reflected in Figure 5.

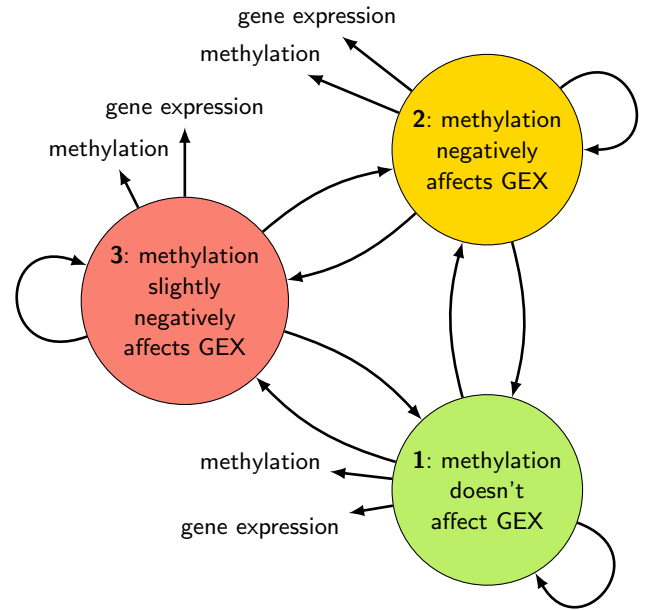


Figure 5 Updated state diagram with new descriptions based on learned parameters

As seen in Figure 6, many genes had similar parameters, regardless of their performance or R^2 value.

For the best gene, the transition matrix was the identity matrix. However, even for the 300th place gene, the transition matrix looked somewhat similar to that of the best gene, demonstrating that state assignments of adjacent CpG sites were largely consistent, hinting as to why the HMM was able to predict gene expression so accurately. While this reveals a biological reality of highly correlated methylation values of nearby CpG sites, it also implies that modeling the sequences of CpG sites for each gene did not have much predictive power.

Best gene transition probabilities matrix:
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

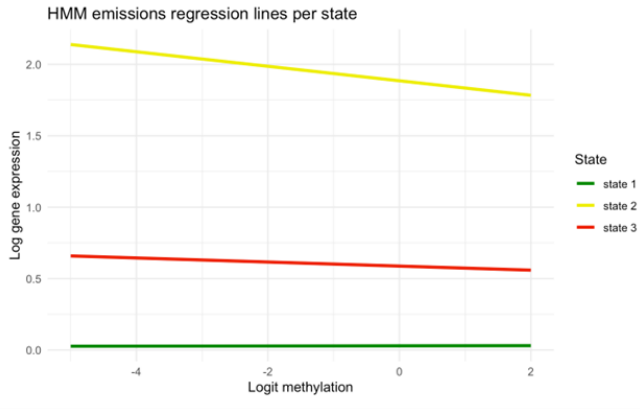


Figure 6 Learned regression models for each state in the HMM

$$300^{\text{th}} \text{ place gene: } \begin{bmatrix} 0.9064 & 0.0936 & 0 \\ 0.2618 & 0.3482 & 0.39 \\ 0 & 0.1007 & 0.8993 \end{bmatrix}$$

4. Conclusion

Based on the comparison between the results from the LASSO analysis and the HMM, it is clear that the HMM significantly outperforms LASSO in predicting gene expression. The HMM can capture the spatial structure of methylation levels of CpG sites across the promoter region of a gene, exemplified by the transition probabilities matrix of the gene with the highest R^2 value being the identity matrix. It shows a consistent assignment of states for adjacent CpGs. However, the strength of modeling CpG sites as a sequence of states that depend on each other was more unclear. It can be inferred, then, that the spatial structure itself does not contribute much predictive power to the HMM's goal of gene expression prediction. Additionally, the HMM struggles to find a strong link between methylation and gene expression in the emission parameters of many of the genes, including the gene with the highest R^2 .

Evidently, the HMM generated a much higher percentage of genes with R^2 greater than 0.3, 0.5, and 0.8, compared to the LASSO. However, based on the transition probabilities and emission parameters of the best gene, the HMM's outperformance of LASSO can most likely be attributed to the state assignments of CpG sites paired with the subsequent calculation of proportions of CpGs in each state to predict gene expression with a linear model. Moreover, the state emissions show that methylation level, rather than methylation-gene expression relationship, played a role in CpG site state assignments. Thus, if methylation level was the primary contributor to state assignments, and state assignments with their subsequent proportion calculations and linear model fit were primary contributors to higher-accuracy gene expression prediction, then by transitivity, methylation level of CpG sites alone is the likely basis for the HMM's high predictive performance.

Furthermore, The HMM was only able to fit 399 out of the 1000 target genes used in LASSO. Future improvements could address some of these limitations by running both the baseline LASSO analysis and the HMM on several thousand more genes to ensure that this trend holds across more of

the genome. Additionally, we could expand the scope of the gene from just the promoter region to analyze if the specific location of CpG sites observed raised any biases in the HMM's learned methylation-gene expression regressions. In this project, cancer data was used since it was the most reliable multi-sample data available and easily accessible, but cancerous cells tend to have abnormal physiology and specifically methylation levels, so rerunning the pipeline on multi-sample healthy tissue would also confirm that this result is not due to the nature of cancer cells (Yu et al., 2017). However, this HMM architecture has potential to better capture the relationship between methylation and gene expression than existing methods.

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The results shown here are in whole or part based upon data generated by the TCGA Research Network: <https://www.cancer.gov/tcga>.

Data availability

The data and code used to generate these results are available in the GitHub repository at <https://github.com/isabellalazarov/methyl-hmm>.